



Relative extents, mechanisms, and kinetics of fluoride removal from synthetic groundwater by prevalent sorbents

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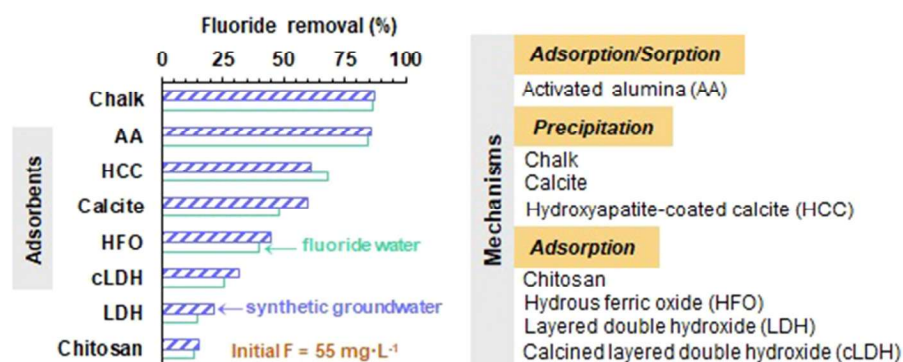
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HIGHLIGHTS

- Defluorination mechanisms and kinetics of eight sorbents compared in groundwater
- Activated alumina and chalk showed the highest F uptake percentages
- The highest sorptive fluoride uptake kinetics was quantified for calcined-LDH
- The highest precipitative fluoride uptake kinetics was quantified for chalk
- Hydroxyapatite-coated calcite was the most promising sorbent for sustainable F removal

GRAPHICAL ABSTRACT



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ABSTRACT

Fluoride (F) contamination in groundwater affects millions of people across the world. Although several sorbents have been identified for low-cost F removal, the choice of the optimal sorbent is dictated by the specific chemistry of contaminated groundwater. In this contribution, eight prevalent sorbents—activated alumina (AA), calcite, hydroxyapatite-coated calcite (HCC), natural chitosan, chalk, Mg–Al–CO₃ layered double hydroxide (LDH), calcined Mg–Al–CO₃ LDH (cLDH), and hydrous ferric oxide (HFO)—were categorized on their relative F removal mechanisms, extents, and kinetics from a typical synthetic groundwater, representative of contaminated aquifers of North India. Initially, batch experiments containing sorbents at 4 g·L⁻¹ were conducted at a high F concentration (2.9 mM). The dominant F removal processes were identified by aqueous- and solid-phase characterization of reaction by-products. While chalk and calcite removed F by secondary precipitation of fluorite, HCC removed F by fluorapatite precipitation, and other sorbents removed F by sorption. Depending on the immobilization mechanism identified, the F uptake kinetics on each sorbent was modeled with either pseudo-

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